Prof. Ashmeet Singh IIT Delhi, Spring 2026
ashmeet@iitd.ac.in

Tuesday, 20/1/2026

---

1 The Interatomic "Morse" Potential [15 points]

The potential energy function is given by,

$$U(x) = D\left(1 - e^{-a(x-b)}\right)^2 . \tag{1}$$

a The units of D are that of energy, i.e., Joule. The units of b are that of distance, i.e. meters. The units of a are that of inverse length, or m^{-1}.

b To find the equilibrium point, it suffices to set $\frac{dU(x)}{dx} = 0$.

$$\implies 2D(1 - e^{-a(x-b)})\; e^{-a(x-b)}(-a) = 0 \tag{2}$$
$$\implies x = b. \tag{3}$$

The equilibrium point is stable. At $x = b$, the potential is 0. On either side, the magnitude of $1 - e^{-a(x-b)}$ increases and the square converts the sign to positive, so $x = b$ is really a minimum and hence is stable.

c The second derivative of $U(x)$ is given by,

$$U''(x) = 2Da^2 e^{-a(x-b)}(2e^{-a(x-b)} - 1). \tag{4}$$

At $x = b$, this is $2Da^2$. The first derivative is of course 0, and U itself is equal to 0 at $x = b$. Hence the morse potential can be expanded around its equilibrium point $x = b$ as,

$$U(x) \approx 0 + 0 + \frac{1}{2!}2Da^2(x-b)^2 = Da^2(x-b)^2 \tag{5}$$

d Since the potential has been converted to a Harmonic approximation, we find the frequecy of oscillations by comparing:

$$\frac{1}{2}m\omega^2(x-b)^2 = Da^2(x-b)^2 \tag{6}$$
$$\implies \omega = a\sqrt{2D/m}. \tag{7}$$

An example of the potential for the parameters $D = 4, a = 1, b = 2$ is shown in Fig. 1.

e

$$U_{LJ}(x) = -\frac{\alpha}{x^6} + \frac{\beta}{x^{12}} . \tag{8}$$

So, first we set $\frac{d}{dx}U_{LJ}(x) = 0$ which gives us $\alpha x^6 = 2\beta \implies x_0 = \left(\frac{2\beta}{\alpha}\right)^{1/6}$ as the point of equilibrium. Then we take the second derivative, which gives us:

$$U''(x) = -42\alpha x^{-8} + 156\beta x^{-14} \tag{9}$$

At the equilibrium position $U''(x_0) = 36\alpha x_0^{-8}$. After substituting for x_0, we get

$$\omega = 6\sqrt{\frac{\alpha^{7/3}}{m(2\beta)^{4/3}}}. \tag{10}$$

We wish to equate this to the Morse-potential frequency. If we do not impose the equilibrium position to be matched, the we get a constraint equation:

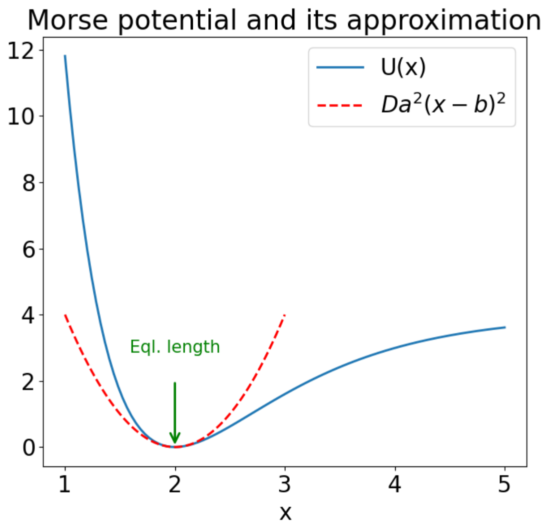


Figure 1: Morse Potential with equilibrium length and curvature.

$$\omega_{LJ} = \omega_M \tag{11}$$

$$\implies \beta = \frac{1}{2}\left(\frac{18}{Da^2}\right)^{3/4} \alpha^{7/4} \tag{12}$$

If now the equilibrium position is matched too, $x_0 = b$ then we get:

$$x_0 = b = \left(\frac{2\beta}{\alpha}\right)^{1/6} \tag{13}$$

This second equation in α and β helps solve the system exactly and gives us the mappings:

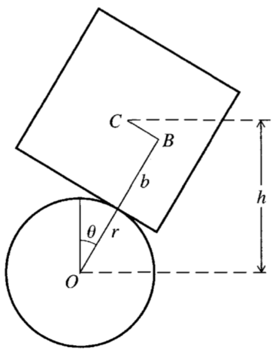
$$\alpha = \frac{Da^2 b^8}{18} \quad , \quad \beta = \frac{Da^2 b^{14}}{36} \tag{14}$$

2 Falling Through the Earth = Orbiting The Earth?

The solution for this problem is reflected in this YouTube video by Prof. Singh:

https://www.youtube.com/watch?v=TI9biu93i4w

3 A Balancing Act



a The block rocks sideways on the cylinder and the position of the COM of the block can be characterized entirely by the angle θ alone. Next, since the block can't slip on the rubber cylinder, this means that the distance of the contact point P from the middle of the block edge is equal to the arc length it has traversed on the cylinder.

Therefore, in the diagram, the distance $BC = r\theta$.

Now, the vertical distance between the centres of mass, h can be written simply as:

$$h(\theta) = (r+b)\cos(\theta) + r\theta\sin(\theta). \tag{15}$$

The potential energy of the block can then be written simply as:

$$U(\theta) = mgh(\theta) \tag{16}$$

b Now, for equilibrium position, we have,

$$U'(\theta) = mg(-(r+b)\sin(\theta) + r\sin(\theta) + r\theta\cos(\theta)) \tag{17}$$

So, now we can clearly see that for $\theta = 0$, we get $U'(\theta) = 0$, an equilibrium.

c Next, to find stability, we should calculate the second derivative of potential energy. That is given by,

$$U''(\theta) = mg\left(-(r+b)\cos(\theta) + r\cos(\theta) + r\cos(\theta) - r\theta\sin(\theta)\right) \tag{18}$$

$$\implies U''(\theta = 0) = mg(-r - b + 2r) = mg(r - b) \tag{19}$$

So, the potential can be expanded around $\theta = 0$ as,

$$U(\theta) \approx mg\left[r + b + \frac{1}{2}(r-b)\theta^2\right] \tag{20}$$

.

So, clearly, the rocking sideways is stable if the radius of the cylinder is bigger than the half-edge length. This inherently makes sense that the the top of the cylinder being flatter (less curvature; curvature is inversely dependent on radius) helps stability.
